Labs

Questions

1. What is resiliency, and why is it essential in distributed systems?
2. Describe the *one thread per request* pattern and its key issues (i.e., thread pool/memory saturation).
3. Which are the most used alternatives to the *one thread per request* pattern? Highlight their key features and mutual differences.
4. Describe the difference between client-side and server-side resiliency patterns.
5. Describe the most used client-side resiliency patterns.
6. Describe the most used server-side resiliency patterns.
7. Describe pros and cons of *fixed-window*, *sliding-window*, *leaky bucket* policies for circuit breaker implementations.
8. What is a fallback mechanism and how does it relate with the circuit breaker pattern?
9. What is Redis persistence, and how does it ensure data durability? Describe the differences between RDB, AOF, and hybrid persistence.
10. Explain how Redis multi-node deployments work. What are the benefits and limitations of using replicas in a Redis setup?
11. Discuss the key caching patterns, and the role of Redis as a caching solution.